

ENNA, Italy

WMO: 164500

Lat: 37.568N

Lon: 14.280E

Elev: 1003

StdP: 89.84

Time Zone: 1.00 (EUC)

Period: 02-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.0	0.2	-13.5	1.3	12.0	-9.2	1.9	11.7	16.5	7.3	13.3	4.9	4.1	300	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	32.5	18.5	30.9	17.9	29.3	17.3	23.1	27.0	22.0	26.0	20.9	24.9	2.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	19.0	24.5	20.9	17.6	23.9	19.8	16.4	22.8	74.5	26.9	69.9	25.3	65.7	24.7	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.6	8.4	7.0	-3.1	35.8	1.5	2.8	-4.1	37.8	-5.0	39.4	-5.8	41.0	-6.9	43.0		
(5)			-4.0	24.2	1.3	2.8	-5.0	26.3	-5.8	27.9	-6.5	29.5	-7.5	31.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	6.4	5.9	8.0	11.6	16.0	21.2	24.5	24.5	19.9	16.3	11.4	8.2		
	DBStd	7.38	3.04	3.17	3.51	3.62	3.80	4.23	2.97	3.20	3.50	3.27	3.10	2.72		
	HDD10.0	432	116	119	84	23	1	0	0	0	1	20	68			
	HDD18.3	1977	369	347	322	202	94	18	1	1	21	81	208	314		
	CDD10.0	2092	5	6	20	72	187	335	450	449	297	196	63	12		
	CDD18.3	596	0	0	0	2	21	103	192	191	68	18	1	0		
	CDH23.3	3797	0	0	0	2	86	635	1379	1354	306	34	1	0		
	CDH26.7	1172	0	0	0	0	14	196	448	444	68	2	0	0		
(14) Wind	WSAvg	2.7	3.0	3.3	3.3	3.1	2.6	2.3	2.1	2.4	2.5	2.5	2.9	2.7		
(15) Precipitation	PrecAvg	439	59	47	46	32	22	11	7	31	37	73	55	60		
	PrecMax	757	240	175	183	104	95	55	61	253	114	213	263	157		
	PrecMin	138	1	5	0	1	0	0	0	0	2	0	0	2		
	PrecStd	171	55	43	43	26	24	16	13	51	26	53	54	46		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.1	16.2	18.8	22.2	28.8	33.4	34.8	34.9	31.1	26.1	20.2	17.2		
		MCWB	8.7	8.9	10.8	13.0	13.9	16.6	20.8	20.0	19.7	15.0	11.5	10.9		
	2%	DB	12.8	13.5	16.1	20.1	25.6	30.9	32.5	32.2	28.5	24.0	18.1	14.8		
		MCWB	6.9	7.7	10.0	11.6	13.3	15.8	18.5	19.2	17.8	14.8	11.5	10.8		
	5%	DB	11.2	11.8	14.2	18.2	23.8	28.9	30.7	30.8	26.6	22.1	16.5	12.6		
		MCWB	7.0	7.5	8.6	10.8	13.3	16.0	17.9	18.5	17.3	14.5	12.0	10.2		
	10%	DB	10.0	10.1	12.9	16.8	21.4	26.9	29.1	29.2	24.9	20.2	15.1	11.1		
		MCWB	7.1	6.8	8.2	10.1	12.6	15.8	17.3	17.8	17.2	14.3	12.1	9.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.1	11.6	14.5	15.5	19.0	22.1	24.8	24.7	23.7	20.0	16.4	13.8		
		MCDB	12.6	13.5	15.5	19.6	21.9	26.6	29.7	28.3	26.5	21.5	17.1	14.8		
	2%	WB	9.8	10.0	12.0	13.9	17.0	20.5	23.0	23.1	22.0	18.4	15.1	12.2		
		MCDB	10.6	11.4	13.6	17.2	20.6	25.0	27.4	27.2	24.7	20.4	15.6	13.3		
	5%	WB	9.0	8.9	10.2	12.8	15.6	19.2	21.6	22.1	20.5	17.4	14.1	11.0		
		MCDB	9.7	10.1	12.3	15.8	19.2	24.1	25.5	26.6	23.0	19.0	14.8	11.8		
	10%	WB	8.2	7.9	9.2	11.7	14.5	17.9	20.4	21.0	19.3	16.6	13.1	10.1		
		MCDB	8.9	9.0	11.2	14.5	18.6	22.9	24.7	25.5	21.8	18.0	13.9	10.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	4.8	5.7	6.5	7.7	8.3	8.6	8.7	7.3	5.9	4.5	4.0		
		MCDBR	5.9	7.1	8.0	8.6	9.3	9.3	9.5	9.5	8.7	8.0	6.2	5.7		
	5% WB	MCWBR	4.8	5.5	6.5	6.3	6.1	5.3	5.6	5.6	5.9	5.7	4.9	4.4		
		MCWBR	4.8	5.8	6.9	7.8	8.2	8.9	9.1	9.3	8.0	6.8	5.1	4.6		
(40) Clear-Sky Solar Irradiance	taub	0.308	0.328	0.367	0.403	0.404	0.387	0.383	0.391	0.391	0.368	0.339	0.314			
		taud	2.473	2.397	2.275	2.183	2.222	2.299	2.315	2.292	2.312	2.366	2.419	2.471		
	Ebn at Noon	879	901	892	877	879	891	891	878	857	843	830	845			
		Edh at Noon	82	100	124	144	141	131	128	118	102	86	77			
(44) All-Sky Solar Radiation	RadAvg	2.24	3.00	4.26	5.46	6.59	7.41	7.68	6.73	5.00	3.63	2.51	2.02			
	RadStd	0.19	0.31	0.35	0.41	0.38	0.36	0.26	0.39	0.25	0.27	0.19	0.18			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	-0.37	-0.47	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page